Muhammad Taha *Electrical Engineer*

tahaansari1703@gmail.com

+92 3273712398

Karachi, Sindh, Pakistan

linkedin.com/in/1aha

github.com/Muhammad-Taha-Ansari

muhammad-taha-ansari.github.io/portfolio

Summary

Electrical Engineering student with hands-on experience in hardware security, digital design, and embedded systems. Proficient in SystemVerilog, C++, and Python with demonstrated expertise in FPGA development and cryptographic protocol implementation including ECDH. Experienced in semiconductor industry environments through internships, with a research background in cybersecurity and computer systems. Strong analytical and problem-solving skills applied to real-world engineering challenges.

Education

NED University of Engineering & Technology , *B.E Electrical Engineering* — 09/2023 – Present, Karachi, Pakistan

Aptech Learning , *Diploma - Web Development* — 10/2023 – 10/2024, Karachi, Pakistan

Meritorious Science College , *Intermediate - Pre-Engineering* — 08/2021 – 08/2023, Karachi, Pakistan

St. Paul's English High School , *Matriculation - Computer Science* — 08/2019 – 05/2021, Karachi, Pakistan

Experience

Xcelerium , *RTL Design & Verification Trainee* — 12/2025 – 06/2026, Karachi, Pakistan

Completed training at Xcelerium as a Training Engineer (RTL/ASIC), focusing on ASIC design, RTL coding, and digital system verification. Worked on SystemVerilog and RTL design, gaining practical experience in industry-standard ASIC design flows including synthesis, timing analysis, and functional verification. Also gained hands-on experience in FPGA-based implementation, including design and implementation of RISC-V processor architecture, strengthening understanding of digital system design.

Neuro-Computational Lab , *Hardware Security Intern* — 02/2026 – 03/2026, Karachi, Pakistan

Completed a one-month internship at NCL, working on Post-Quantum Cryptography and Hardware Security with a focus on Elliptic Curve Diffie-Hellman (ECDH). Gained hands-on exposure to cryptographic algorithm implementation, hardware security principles, and emerging post-quantum cryptographic techniques aimed at securing systems against quantum computing threats.

NAT Solar Energy 🔗, *Solar System Installation Intern* 03/2025 – 04/2025

Karachi, Pakistan

Completed a one-month internship at NAT Solar Energy, where I assisted in site surveying, installation, and wiring of solar PV systems while ensuring compliance with safety regulations and system design standards. Performed basic troubleshooting and maintenance to identify faults and ensure optimal system performance, and participated in client meetings to understand energy requirements and recommend suitable energy-saving solar solutions.

Projects

ML - KEM (Kyber), *Hardware Security | Post Quantum Cryptography* 06/2026 – Present

- Implementing hardware architecture for NIST FIPS 203-standardized ML-KEM (Kyber) post-quantum cryptographic algorithm using RTL design in SystemVerilog.
- Researching Number Theoretic Transform (NTT)-based polynomial arithmetic and lattice-based key encapsulation mechanisms (KEM) to develop an area- and power-efficient FPGA-targeted design.
- Analyzing CRYSTALS-Kyber specification to extract hardware-level design constraints, supporting secure key generation, encapsulation, and decapsulation module design.

Elliptic Curve Diffie-Hellman (ECDH) 🔗, *Hardware Security | Post Quantum Cryptography* 02/2026 – 03/2026

- Designed and implemented a hardware accelerator for Elliptic Curve Diffie-Hellman (ECDH) secure key exchange protocol using RTL design in SystemVerilog.
- Developed and verified a fully functional RTL design with simulation testbenches, validating correctness of elliptic curve point multiplication and shared secret generation.

GENIC – AI Humanoid Robot 🔗, *Artificial Intelligence* 06/2026 – Present

- Independently designed and developed GENIC, a full-stack AI-powered humanoid robot platform integrating computer vision, NLP-based voice command recognition, and machine learning on an ESP32 microcontroller.
- Built real-time object and face detection pipeline using Python and OpenCV, enabling the robot to perceive and respond to its environment autonomously.

Risc-V Processor 🔗, *VSLI - Digital Design* 11/2025 – 06/2026

- Designed and implemented three variants of a RISC-V processor in Verilog/SystemVerilog, each integrating a distinct communication interface - SPI, UART, and I2C to support peripheral connectivity.
- Verified all processor variants through RTL simulation, validating instruction execution pipeline and protocol-level data transfer across each interface.

Smart Energy Meter (FPGA) 🔗, *VSLI - Digital Design* 02/2026 – 04/2026

- Designed a Smart Energy Meter on DE1-SoC (Intel Cyclone V SoC) integrating FPGA fabric with an ARM Cortex-A9 processor, interfacing an LTC2308 12-bit SPI ADC and ACS711 Hall-effect current sensor for real-time AC power monitoring.
- Built a complete analog front-end using a voltage divider for AC signal scaling and breadboard sensor circuitry, with on-board 7-segment displays for live energy output.

Humidity and Temperature Sensor 🔗, *Embedded Engineer* 03/2026 – 05/2026

- Developed an embedded environmental monitoring system on Arduino (AVR) interfacing an AHT20 digital temperature and humidity sensor via I2C protocol for real-time atmospheric data acquisition.
- Implemented firmware in C/C++ for I2C sensor communication, data parsing, and real-time display output on an OLED/LCD screen, delivering a fully functional standalone embedded instrument.

Digital True RMS Voltmeter 🔗, *Embedded Engineer* 04/2026 – 05/2026

- Designed and implemented a Digital True RMS Voltmeter on an Arduino (AVR) microcontroller, using onboard ADC sampling with firmware-based RMS computation for accurate AC voltage measurement.
- Developed dual output interfaces - I2C-driven OLED display for local readout and UART serial transmission for real-time terminal monitoring - delivering a fully functional embedded measurement instrument.

Ventex Crypto Dashboard 🔗, *Full Stack WebDevelopment* 11/2025 – 12/2025

- Collaborated in a 3-member team to build Ventex Crypto, a full-stack cryptocurrency platform using React.js (frontend), MongoDB (database), and a RESTful backend architecture, deployed on Vercel.
- Responsible for [your role: e.g. frontend UI / REST API development / database schema design], implementing responsive design and real-time data handling for a seamless user experience.

SAP Society Mock Entry Test Platform 🔗, *Full Stack WebDevelopment* 06/2026 – 06/2026

- Built a full-stack university entry test system (Node.js, Express, SQLite) with student registration, payment receipt verification, and auto-generated admit cards with unique test passwords.
- Developed a secure, hidden timed test module (100 MCQs, 2 hours) with per-student shuffled questions, JWT-protected admin authentication, and program-wise subject routing (Pre-Medical, Pre-Engineering, CS).
- Implemented an admin dashboard for payment verification, result analytics, and question bank management with manual entry and CSV bulk import support.

Skills

Programming Languages
C++, Python, JavaScript, HTML, CSS, React

EDA & Simulation Tools
ModelSim, Verilator, Proteus, MultiSim, MATLAB, AutoCAD, ETAP, DIALux

Soft / Leadership Skills
Problem Solving, Debuging, Logical Reasoning, Leadership, Team Management, Event Management, Cross-functional Collaboration, Client/Stakeholder Communication

Hardware Description & Verification
SystemVerilog, Verilog, SystemVerilog Verification, Digital Logic Design, VLSI

Specialized Domains
Object Oreinted Program (OOPs), Post-Quantum Cryptography, Hardware Security, Computer Networks Design, Large Language Models (LLMs), AI Systems (Basic)

Certificates

- GPU Design in SystemVerilog

Organisations

Study Aid Project - NED, *IT & Competition Lead* — 02/2025 – Present, Karachi, Pakistan

Joined SAP Society in February 2025 as a general member and was promoted to IT & Competition Lead based on performance. Manage and develop the society's web platforms and technical infrastructure as IT Lead. Organize and oversee competitions, coordinate with participants, and lead teamwork initiatives as Competition Lead, creating opportunities for students to showcase their skills. Strengthened leadership, collaboration, and event management abilities through this role.

Anjuman Hayat-ul-Islam, *Volunteer* — 02/2025, Karachi, Pakistan

Earned a Community Service Certificate for volunteering at an orphanage, spending three days engaging with children through story telling, creative play, and interactive learning activities. Fostered a joyful and nurturing environment, promoting social and emotional development while demonstrating patience, empathy, & teamwork.

Courses

Cryptography 1, *Coursera* — 06/2026 – Present

Crash Course on Embedded C Programming, *Coursera*

Fundamentals of Digital Design for VLSI Chip Design, *Coursera*

VLSI Course, *Simplilearn - Skillup*

PCB Design Course, *Simplilearn - Skillup*

Introduction to Large Language Models, *Google Cloud*

MATLAB Onramp , *MATLAB@UCL*

Responsive Web Design , *FreeCodeCamp*

Languages

Urdu	●●●●●	English	●●●●○
French	●○○○○	Arabic	●○○○○

Interests

- VLSI - Digital Design
- Hardware Security
- Embedded Systems
- Artificial Intelligence
- Post Quantum Cryptography
- Web Development